Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

Gravity bends a soccer ball's path, buoyancy acts upward, drag reduces its speed, and Magnus lift redirects a spinning ball. We formulate a three-dimensional model including buoyancy and compare four models using existing three-force calculations: gravity alone, constant drag, constant drag and lift, and complete speed- and spin-dependent aerodynamic coefficients. Published Teamgeist relations reproduce the selected 2007 wind-tunnel drag and lift curves with coefficient RMSE values of 0.0113 and 0.0504.

Forward calculations are compared with controlled launch experiments, including a measured range of 21.9 m and four rotating-ball trajectory segments. The variable-coefficient model predicts a range of 21.871 m in the nonspinning experiment. Inverse calculations estimate initial motion and nonnegative force corrections from two later positions, selecting corrections close to unity. Five recorded match flights provide subsequent positions for comparison. For the competition data, the variable-coefficient model predicts the goal-plane centre $(y, z) = (2.683, 1.959)$ m at $t = 0.4026$ s. A separate same-initial-state calculation quantifies rotational damping, while surface geometry and wind are discussed through published flow results and force estimates.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and rotation can produce a sideways or vertical aerodynamic force. Bray and Kerwin combined aerodynamic forces with equations of motion and compared the resulting trajectories with experimental free kicks [5]. Myers and Mitchell analyzed three-dimensional spinning-ball motion and the influence of drag and spin [15]; Maponi and Piombin developed a numerical framework that also accommodates wind [12].

The aerodynamic behavior is also affected by the ball's surface. Wind-tunnel measurements by Asai et al. examined the relationship between flow conditions and soccer-ball forces [16], while Paudel and Yan's simulations compared a football with realistic seam geometry against a smooth sphere [14]. These studies motivate a spherical geometric approximation together with an examination of how aerodynamic coefficients affect the predicted flight.

Measurements of real flights provide a basis for evaluating such models. Griffiths et al. tracked the three-dimensional motion and spin of a marked football in field conditions [3]. Goff and Carré used camera-recorded trajectories to estimate aerodynamic coefficients [20]; subsequent work by Goff et al. combined measurements from 29 flights to construct speed-dependent drag and lift curves [4].

In this problem, the initial position and two early positions are given, whereas the initial velocity and spin are not directly specified. We therefore use the equations of motion to find states compatible with the early observations and calculate their goal-plane crossings. The DFL open dataset combines match events and optical tracking data for professional matches [10, 11], allowing predicted crossings to be compared with later recorded positions.

1.2 Problem Restatement

We model a soccer ball travelling in three dimensions from the vicinity of the penalty mark to the goal. The goal is 7.32 m wide and 2.44 m high; the penalty mark is 11 m from the goal line. We establish a model incorporating gravity, buoyancy, drag and the Magnus effect, solve for its trajectory and the point at which it crosses the goal plane, and compare the calculated positions with a recorded penalty flight. The prediction uses the initial position and the positions at 0.05 and 0.10 s; subsequent positions are reserved for validation. Figure 1 summarizes the force models and inverse problem: early positions determine initial conditions and force corrections, while later positions provide validation targets. Section 2 states the assumptions and notation. Section 3 establishes the forces and equations. Section 4 compares forward and inverse predictions. Section 5 is reserved for sensitivity analysis. Section 6 discusses surface geometry, spin damping and wind, followed by conclusions and model evaluation in Sections 7 and 8.

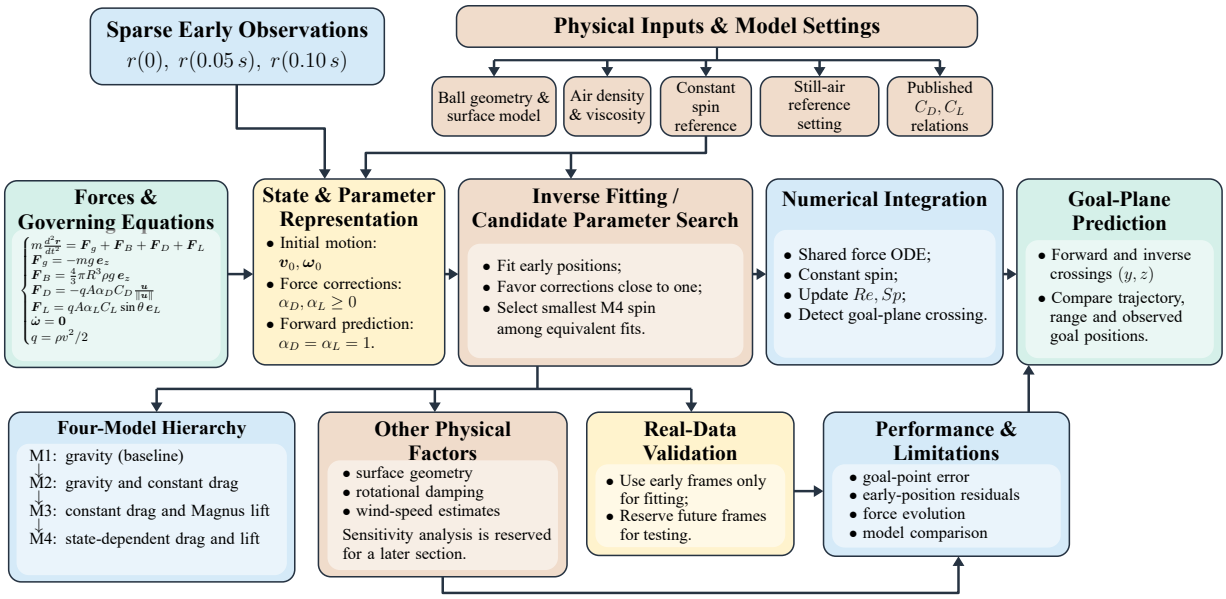


Figure 1: Force model including buoyancy, four-model baseline comparison and goal-plane validation.

2 Preparation of the Models

2.1 Assumptions

- **Ball–foot interaction.** Free flight begins at $t = 0$ immediately after the ball leaves the player's foot. The impact process and deformation during contact are not modeled.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [12, 5, 15]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, air buoyancy, aerodynamic drag and spin-induced lift act during free flight [12, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.
- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.

- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknown parameters.** The position coordinates supplied in the problem are treated as exact. The initial position and the next two early positions are the inverse input; subsequent DFL tracking positions are reserved for comparison [10, 11]. For bounded uncertainty analysis, unresolved initial velocity, spin, aerodynamic parameters and environmental conditions are constrained to stated physically reasonable ranges. Prediction uncertainty is attributed to these unresolved quantities and model assumptions rather than measurement noise. The recorded DFL vertical coordinate is interpreted as height in metres.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [12]; the air density follows Asai et al. [16]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. Controlled experiments use their reported mass, diameter and air properties.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

For a pitch 105 m long with its origin at the centre, the two goal lines are located at $x = \pm 52.50$ m. The selected kick travels towards the positive-x goal. For the competition geometry the valid goal opening is $-3.66 \le y \le 3.66$ m and $0 \le z \le 2.44$ m. The goal-plane position is the ball centre's first forward intersection with $x = X_g$.

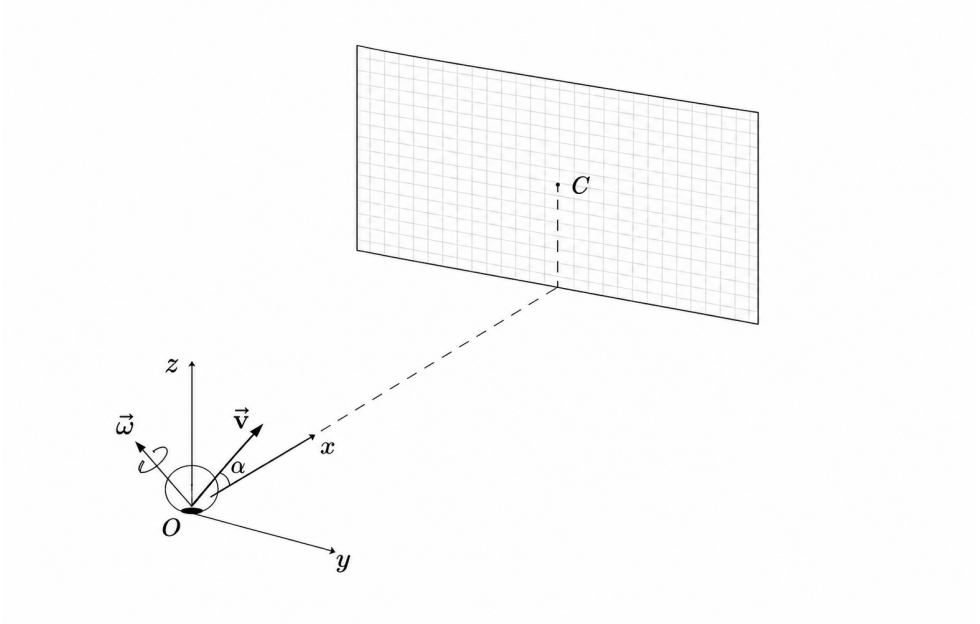


Figure 2: Penalty-kick geometry in local coordinates. O is the penalty mark and C is an illustrative point on the goal plane.

Figure 2 uses the penalty mark as the local origin and places the goal plane at $x = 11$ m. For the match record below, the pitch-centred coordinate system instead places the goal at $X_g = 52.50$ m.

The ball crosses the opening cleanly when its centre satisfies $|y| \le 3.66 - R$ and $R \le z \le 2.44 - R$ at $x = X_g$. Ground contact occurs when the descending centre reaches $z = R$. These event definitions are used consistently for all four models.

3.2 Force Analysis

3.2.1 Air Buoyancy

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{1}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b) g \boldsymbol{e}_z$. For the reference parameters in the table, buoyancy is about 1.56% of the ball's weight and changes the downward acceleration from g to $g(1 - \rho V_b/m)$.

3.2.2 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{2}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.

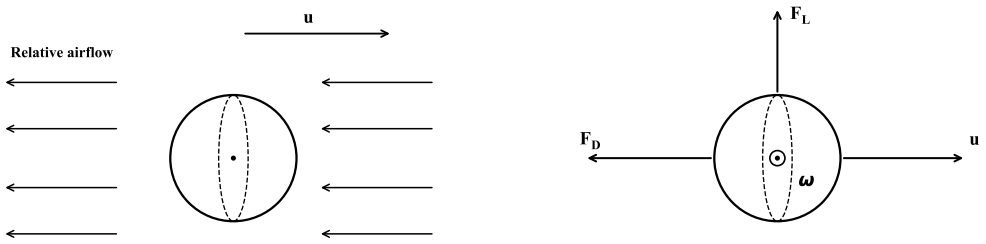


Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.3 Aerodynamic Drag

In the ball frame, the far-field air velocity is $-\boldsymbol{u}$. Approximate the maximum cross-section by a disk of area $A = \pi R^2$ facing the oncoming air. For a uniform stream decelerated from speed v to rest, Bernoulli's relation gives the stagnation-to-far-field pressure difference and a corresponding force scale,

$$\Delta p_{\text{stag}} = \tfrac{1}{2}\rho v^2, \qquad F_{\text{scale}} = \Delta p_{\text{stag}} A = \tfrac{1}{2}\rho A v^2. \tag{3}$$

Pressure and viscous shear vary across the actual ball, so the dimensionless coefficient C_D expresses the integrated drag relative to this scale. Drag opposes the velocity of the ball relative to the air, giving [20, 17]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A \alpha_D C_D(Re, Sp) v \boldsymbol{u}, \qquad \boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}, \quad v = \|\boldsymbol{u}\|. \tag{4}$$

Its quantities have the following physical roles:

1. **Air density $\boldsymbol{\rho}$.** The density measures air mass per unit volume. We use the uniform reference value 1.2 kg/m^3 reported by Asai et al. [16].

2. **Projected area $\boldsymbol{A = \pi R^2}$.** A larger radius intercepts more airflow; the spherical geometry fixes this area throughout flight.

3. **Relative speed and direction $\boldsymbol{v}, \boldsymbol{u}$.** The quadratic speed dependence gives stronger resistance at higher speed, and the negative direction opposes airflow. The still-air main model has $\boldsymbol{u} = \boldsymbol{v}$.

4. **Drag coefficient $C_D(Re, Sp)$.** The coefficient represents boundary-layer transition, separation and wake formation. Following Asai et al. and Kiratidis and Leinweber [16, 17], we define

$$Re = \frac{2Rv}{\nu}, \qquad Sp = \frac{R\omega}{v}, \qquad \omega = \|\boldsymbol{\omega}\|, \quad \nu = \frac{\mu}{\rho}. \tag{5}$$

Re compares inertial and viscous flow effects; Sp compares rotating surface speed with airspeed. Constant spin makes Sp rise as the ball slows.

The drag crisis is the rapid reduction of drag associated with boundary-layer transition. A useful coefficient law should describe this transition as well as the slower variation in the supercritical regime. According to Kiratidis and Leinweber [17], their Eq. (5) represents zero-spin drag by

$$\begin{aligned} f(v) = {} & \frac{a - b_{\text{min}}}{1 + \exp[(v - v_c)/v_s]} + b_{\text{min}} \\ & + \frac{v - v_{\text{min}}}{1 + \exp[-(v - v_{\text{min}})/v_s]} \frac{b_{\text{max}} - b_{\text{min}}}{v_{\text{max}} - v_{\text{min}}}. \end{aligned} \tag{6}$$

The coefficient a sets the low-speed level, v_c locates the transition, and v_s controls its width. The remaining parameters set the level and slope of the high-speed tail. Table 2 gives the Teamgeist parameters reported in their Table 1 [17].

Table 2: Teamgeist zero-spin drag parameters from Kiratidis and Leinweber [17].

a	v_c	v_s	b_{min}	b_{max}	v_{min}	v_{max}
—	m/s	m/s	—	—	m/s	m/s
0.4927	12.58	1.071	0.1440	0.1540	23.17	35.00

For a spinning Teamgeist, Kiratidis and Leinweber's Eq. (7) adds a linear spin correction with slope 0.514, fitted to the supercritical wind-tunnel data [17]:

$$C_D(Re, Sp) = f\left(\frac{Re\nu}{2R}\right) + 0.514Sp. \tag{7}$$

Thus drag changes with speed even at zero spin, and also increases with rotation. The Teamgeist parameter set is used throughout the coefficient comparison; other ball surfaces require their own parameters.

The final M4 trajectory has speeds of 24.99–32.57 m/s and Reynolds numbers of $(3.67–4.78) \times 10^5$. Kiratidis and Leinweber's Teamgeist fit extends to 35 m/s [17], corresponding to $Re = 5.13 \times 10^5$ for the reference air. The competition flight lies in this speed range and in the supercritical regime represented by the empirical relation, supporting its use here. The rotating Asai measurements span $Re = 333793–455172$; the coefficient figures use the measured upper condition exactly. The inferred competition spin gives $Sp = 0.427–0.557$; its higher values extend beyond the selected wind-tunnel spin interval.

5. **Drag correction $\alpha_D \geq 0$.** This dimensionless multiplier equals one in forward calculations and is estimated from early positions in inverse calculations.

Comparison with published wind-tunnel data. Asai et al. [16] measured soccer-ball forces in Tokai University's low-speed recirculating wind tunnel using a six-component balance (1.5×1 m test section; about 1% turbulence). We use their spinning full-size Teamgeist FRP replica: $R = 0.11$ m, $\rho = 1.2$ kg/m^3, airspeed 22–30 m/s and spin 2–10 rev/s. Five points per coefficient are digitized from their Figures 8 and 9 at $Re = 455172$ (about 30 m/s); the reported side force C_S represents transverse Magnus lift here. The speed–Re correspondence gives $\nu \simeq 1.45 \times 10^{-5}$ m^2/s for this matched comparison.

Figure 4 reproduces the increase of drag with Sp: the coefficient MAE is 0.00985 and RMSE is 0.01127. Both curves use the same experimental condition and measured spin interval.

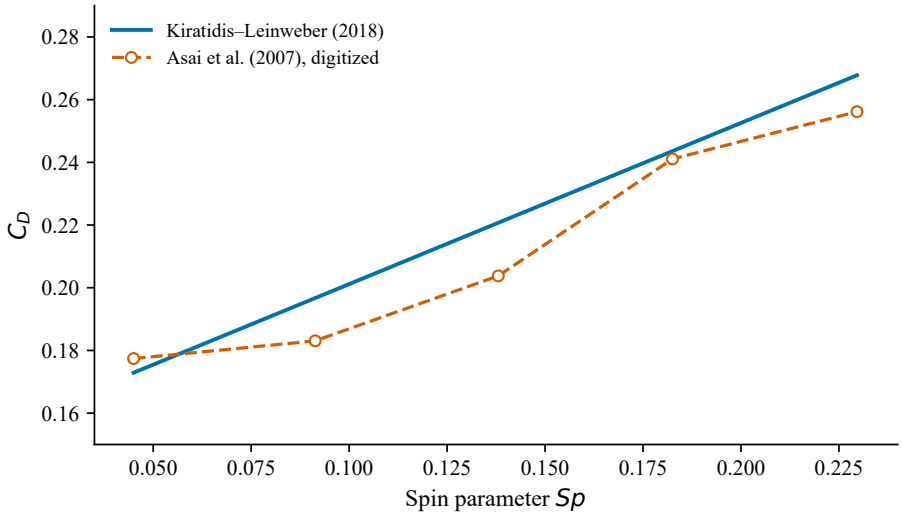


Figure 4: Teamgeist drag at $Re = 455172$: the complete empirical relation [17] and five digitized Asai measurements [16], Figure 8.

3.2.4 Magnus Lift

Figure 5 illustrates relative streamlines in the ball frame. For the spin shown, the upper surface moves with the relative airflow while the lower surface moves against it. Rotation makes the flow asymmetric and produces a transverse force. The figure uses a two-dimensional ideal flow with circulation to illustrate the directions.

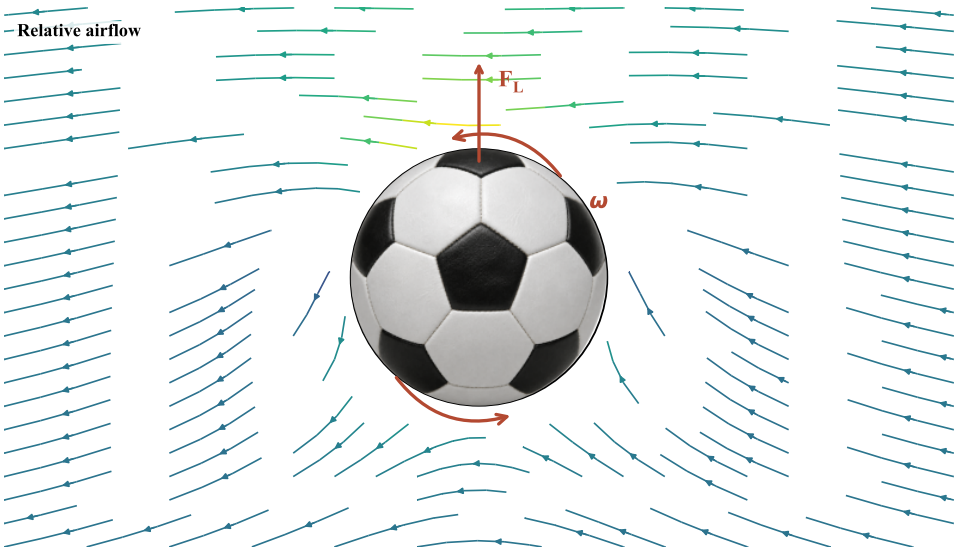


Figure 5: Idealized relative streamlines around a spinning football. Arrows show the directions of relative airflow, spin and Magnus force; colours indicate relative speed in this illustrative flow field.

Rotation changes the flow on opposite sides of the ball and creates a transverse pressure force. Using the drag dynamic-pressure scale gives $F_L = qAC_L$ for transverse spin [20, 17]. For a three-dimensional spin axis, we adopt

$$\begin{aligned} \boldsymbol{F}_L &= \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) v^2 \sin\theta \, \boldsymbol{e}_L, \\ \boldsymbol{e}_L &= \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}, \qquad \sin\theta = \frac{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}{\omega v}. \end{aligned} \tag{8}$$

Lift is zero for zero spin, zero airspeed or parallel spin and airflow.

For $\omega > 0$ and $v > 0$, substituting the definitions of direction and angle into Eq. (8) yields the equivalent form

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) \frac{v}{\omega} (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{9}$$

This form makes the direction perpendicular to both spin and relative velocity explicit. Since C_L also depends on spin, the factor $1/\omega$ does not imply divergent lift at zero spin; zero spin or zero relative speed is handled separately by setting lift to zero.

1. **Dynamic-pressure and area scale $\boldsymbol{\rho A v^2/2}$.** Density, projected area and speed have the same meanings as in the drag force.

2. **Magnus direction $\boldsymbol{e_L}$.** This unit vector is perpendicular to airflow. Topspin bends the path downward, backspin upward, and sidespin horizontally.

3. **Angular projection $\boldsymbol{\sin\theta}$.** The transverse component is one for perpendicular spin and airflow and vanishes for parallel directions. This projection extends the transverse empirical law to an oblique spin axis.

4. **Lift coefficient $C_L(Re, Sp)$.** Rotation and boundary-layer state jointly determine lift.

   Rotation creates an asymmetric flow and a transverse Magnus force. Its magnitude depends on the boundary-layer state as well as spin, so a spin-only relation does not describe all flow conditions. Kiratidis and Leinweber [17] introduce the reference power law in their Eq. (8),

   $$C_L^{\text{fit}}(Sp) = 1.15 Sp^{0.83}, \tag{10}$$

   and scale it using drag as an indicator of flow state in their Eq. (9) [17]. We use that complete relation:

   $$C_L(Re, Sp) = C_L^{\text{fit}}(Sp) \frac{C_{D,\text{ceil}} - \min[C_D(Re, Sp), C_{D,\text{ceil}}]}{C_{D,\text{ceil}} - C_{D,\text{ref}}}, \tag{11}$$

   with the Teamgeist reference quantities

   $$C_{D,\text{ceil}} = f(0), \qquad C_{D,\text{ref}} = C_D(333793, 0.19). \tag{12}$$

   Here $C_{D,\text{ceil}} \simeq 0.4927$ is the zero-spin low-speed ceiling. The reference drag is evaluated with Eq. (7), including its spin term and using the same R and ν as the calculation. At the reference $(Re, Sp) = (333793, 0.19)$ the scaling factor is one and $C_L \simeq 0.290$. The ceiling, reference condition and use of drag to reduce lift follow Kiratidis and Leinweber [17]; Eq. (11) supplies the final lift coefficient in the force model.

5. **Lift correction $\alpha_L \geq 0$.** This dimensionless multiplier is one in forward prediction and is fitted to early positions when inverse prediction includes lift.

Comparison with published wind-tunnel data. Figure 6 uses the same Asai experiment and flow condition. The complete relation reproduces increasing lift with Sp and lies above all five measured values: MAE 0.04505, RMSE 0.05040 and maximum difference 0.07325. This larger discrepancy motivates the inverse lift correction α_L.

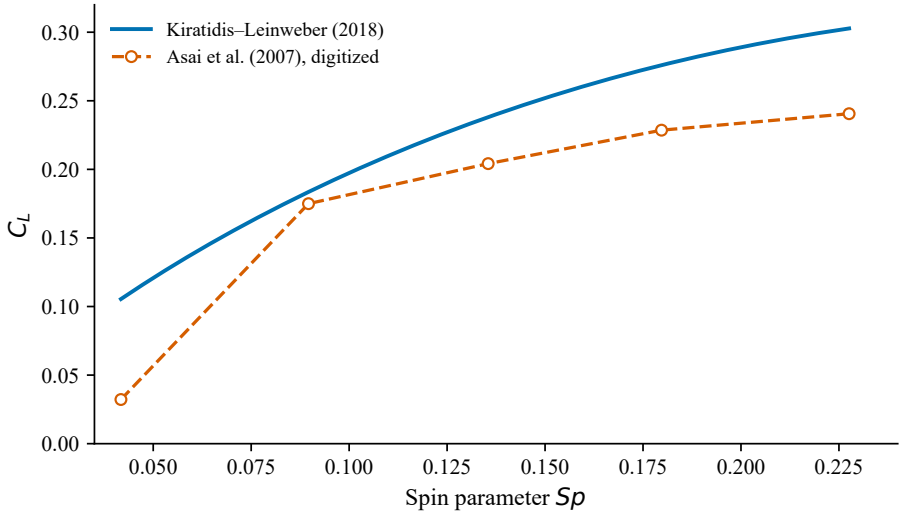


Figure 6: Teamgeist transverse lift at $Re = 455172$: the complete empirical relation [17] and five digitized Asai side-force measurements [16], Figure 9.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{13}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_B + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{14}$$

Substituting drag and Eq. (9) into Newton's second law gives the components used for numerical integration. Define $k_D = \rho A \alpha_D C_D v/(2m)$ and $k_L = \rho A \alpha_L C_L v/(2m\omega)$. For $\omega > 0$ and $v > 0$,

$$\begin{aligned} \dot{v}_x &= -k_D u_x + k_L(\omega_y u_z - \omega_z u_y), \\ \dot{v}_y &= -k_D u_y + k_L(\omega_z u_x - \omega_x u_z), \\ \dot{v}_z &= -g(1 - \rho V_b/m) - k_D u_z + k_L(\omega_x u_y - \omega_y u_x). \end{aligned} \tag{15}$$

The components are integrated together with $\dot{\boldsymbol{r}} = \boldsymbol{v}$; at zero spin or relative speed the lift components are set to zero.

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{16}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

This section presents the equations extended to include buoyancy. The existing numerical results and figures in Section 4 and later sections were calculated using gravity, drag and Magnus lift, corresponding to the baseline $\boldsymbol{F}_B = \boldsymbol{0}$; the buoyancy-inclusive results have not been recomputed.

4 Model Solving

The following examples retain the existing three-force baseline, without buoyancy. We compare increasing levels of aerodynamic description to identify which effects control the trajectory. Table 3 defines the four models. Forward prediction uses measured or reconstructed launch conditions and unit force corrections. Inverse prediction uses two later positions to estimate the initial motion and force corrections, then propagates the fitted state to the goal.

Table 3: Four-model hierarchy. Gravity and constant spin are common to all models.

Model	Drag	Magnus lift	Inverse quantities
M1	none	none	$\boldsymbol{v}_0$ (3)
M2	constant C_D^*	none	$\boldsymbol{v}_0, \alpha_D$ (4)
M3	constant C_D^*	constant C_L^*	$\boldsymbol{v}_0$, spin axis, α_D, α_L (7)
M4	$C_D(Re, Sp)$	$C_L(Re, Sp)$	$\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L$ (8)

For M2 and M3, the constants are evaluated once at the published reference point:

$$C_D^* = C_D(333793, 0.19), \qquad C_L^* = C_L(333793, 0.19) = 1.15(0.19)^{0.83}. \tag{17}$$

For the competition air and ball parameters, $C_D^* = 0.2415$ and $C_L^* = 0.2898$. All models use the same physical conditions within each comparison. M1 and M2 have no spin-dependent trajectory; M3 uses the spin-axis direction, and M4 uses both its direction and magnitude.

4.1 Forward Prediction

4.1.1 Prediction procedure and experimental conditions

Given $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$, we set $\alpha_D = \alpha_L = 1$ and integrate Eq. (14) using an adaptive Runge–Kutta method. A root of $x - X_g$ locates the goal-plane crossing, and a descending root of $z - R$ locates ground contact. The calculation records speed and aerodynamic force magnitudes along the flight.

We compare five controlled flights: one nonspinning launch, with a measured range of 21.9 m, and four topspin trajectories. Their initial states are listed in Table 4.

> **Data source and processing.** Goff and Carré [20] used a 32-panel ball (mass 0.424 kg; diameter 0.218 m) and two high-speed cameras, reporting air density 1.2 kg/m^3, kinematic viscosity 1.54×10^{-5} m^2/s and landing-range error at most ± 0.05 m. We extract the red markers in their Figures 3, 4, 11 and 14 from the vector PDF. The nonspinning case uses approximately 18 m/s, 22° and 0.85 m initial height; its early marker spacing is 0.005 s. For the four spinning cases, published Re and Sp determine speed and spin, and the first three extracted positions determine the initial spatial tangent; comparison uses the spatial curve $z(x)$.

Table 4 specifies the forward inputs. All examples lie in the x–z plane, with $y_0 = v_{y0} = 0$ and $\omega_{x0} = \omega_{z0} = 0$; the listed v_{x0}, v_{z0} and ω_{y0} therefore complete the six initial velocity and spin components. Each first marker fixes the initial position.

Table 4: Initial conditions for the controlled forward examples, reconstructed as described above.

Launch	x_0 (m)	z_0 (m)	v_{x0} (m/s)	v_{z0} (m/s)	ω_{y0} (rad/s)
Zero spin	0.0001	0.8501	16.689	6.743	0.00
Topspin a	0.0001	0.8499	16.825	7.366	82.57
Topspin b	0.0001	0.8499	18.194	6.810	44.56
Topspin c	-0.0070	0.8526	17.383	8.188	125.16
Topspin d	-0.0067	0.8500	18.770	10.863	47.75

4.1.2 Calculated trajectories and physical interpretation

Table 5: Nonspinning controlled launch: predicted landing and signed range error. The reported range is 21.9 m.

Model	Range (m)	Prediction minus measurement (m)
M1	24.650	+2.750
M2	20.656	-1.244
M3	20.656	-1.244
M4	21.871	-0.029

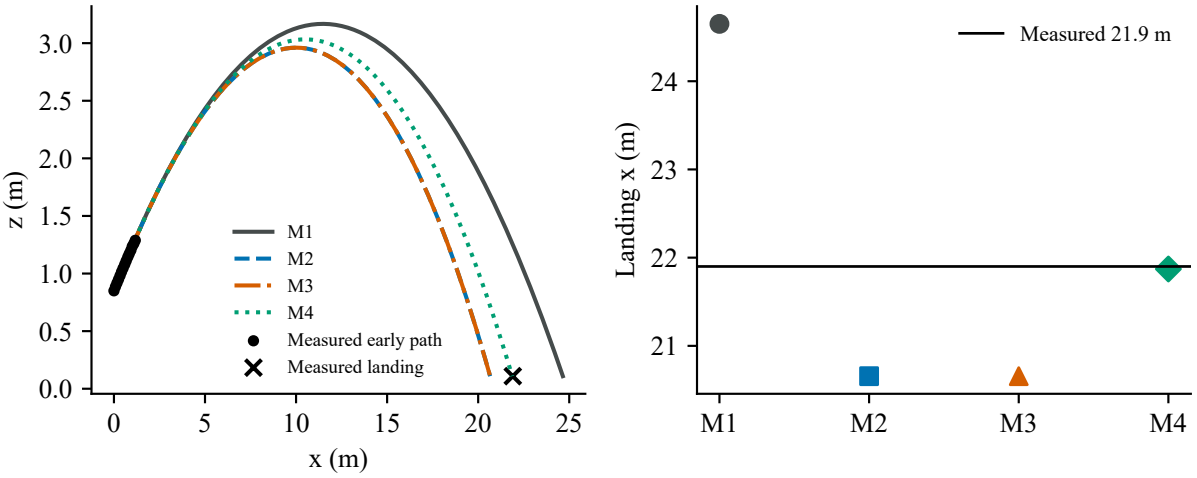


Figure 7: Forward prediction of Goff and Carré's nonspinning experiment [20]. Left: calculated flight paths, digitized early positions and measured landing. Right: individual model ranges against the measured 21.9 m. M2 and M3 coincide because the initial spin is zero.

Gravity alone produces the longest range. Air resistance removes horizontal momentum and shortens flight distance. M2 and M3 use the same constant drag and predict identical nonspinning trajectories. M4 allows drag to change with the slowing airflow; its predicted range differs from the measured value by -0.029 m under the stated approximate launch conditions.

Table 6: Height RMSE (m) at digitized horizontal positions for the four topspin segments.

Launch	M1	M2	M3	M4
a	0.0119	0.0118	0.0062	0.0072
b	0.0396	0.0397	0.0459	0.0466
c	0.0100	0.0100	0.0047	0.0099
d	0.0171	0.0170	0.0090	0.0082

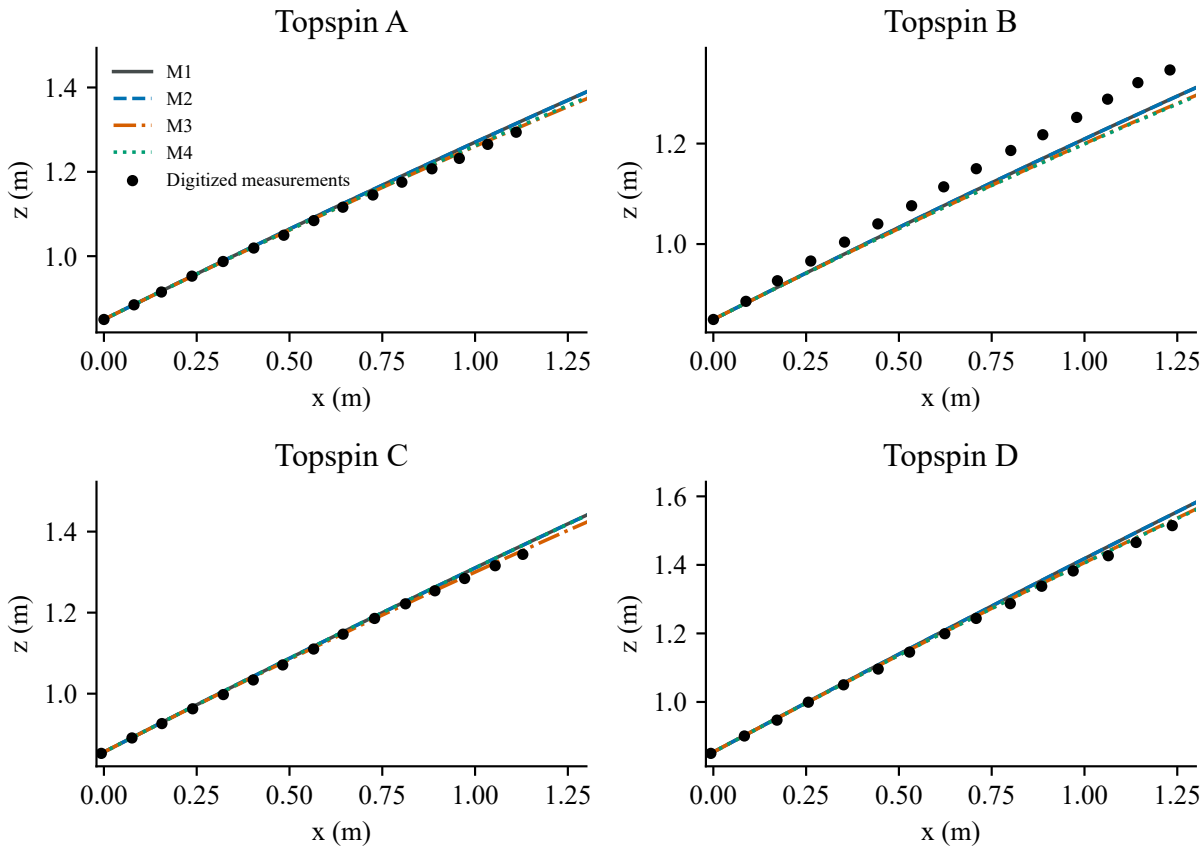


Figure 8: Four topspin launches digitized from Goff and Carré [20], Figure 14. Black points are measured marker centres; model curves use unit force corrections and the stated initial conditions. Each panel covers the published early spatial segment.

Topspin produces downward Magnus force and increases downward curvature. Table 6 gives the height RMSE at measured horizontal positions. The ranking varies across launches. Local direction estimates and the experimental ball's surface both influence the comparison with the Teamgeist reference. In topspin c, the high spin parameter raises the empirical drag above the ceiling in Eq. (11), so M4 assigns zero lift. The measured path retains downward spin-induced curvature, and M3 consequently gives a smaller local error in that case. These examples connect differences in the coefficient law to differences in the predicted path.

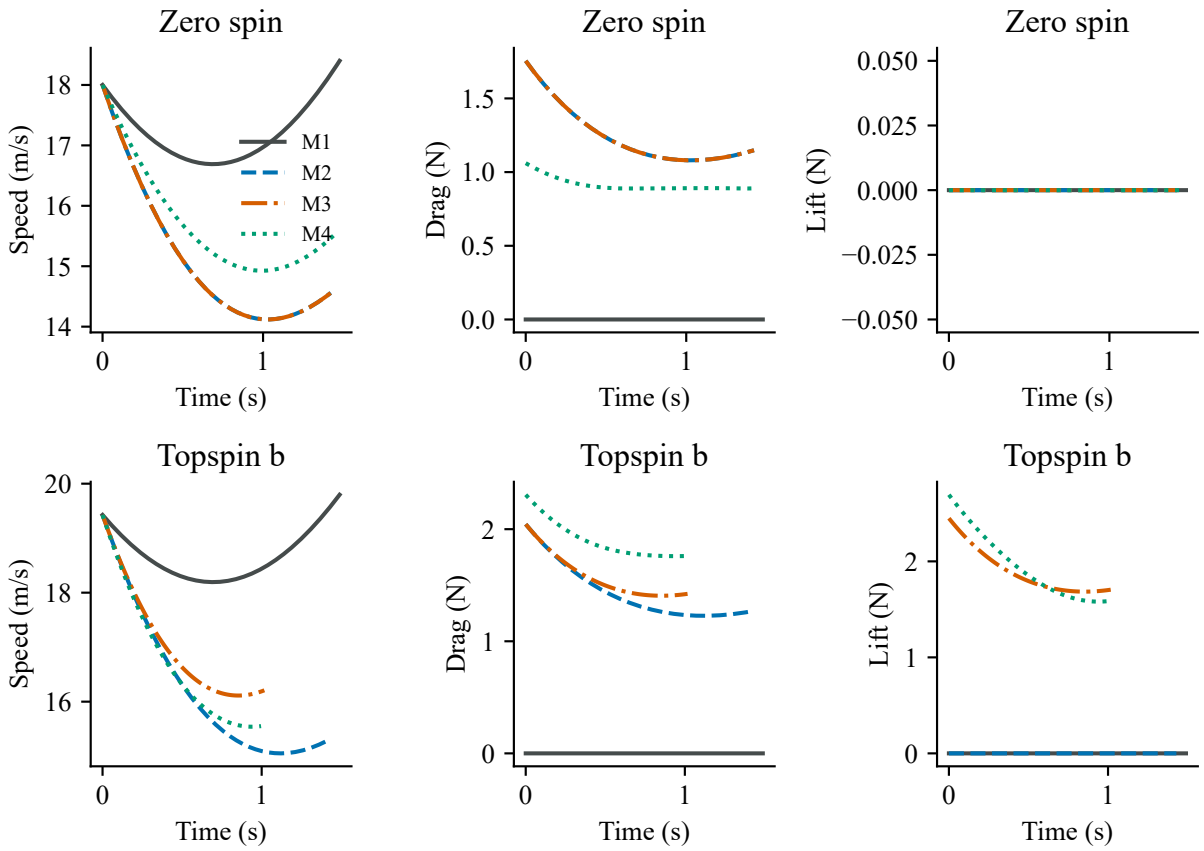


Figure 9: Speed, drag and lift during the nonspinning and topspin-b forward examples. Force magnitudes are calculated from the shared equations. Constant aerodynamic coefficients still produce changing forces because dynamic pressure varies with speed.

Figure 9 shows the mechanism behind these paths. Drag is strongest when the ball is fast. Downward lift in a topspin flight changes its vertical motion and flight duration. In M4, changing speed also changes Re and Sp, so the force evolution includes both dynamic-pressure and coefficient effects.

4.2 Inverse Prediction

4.2.1 Early-position fitting and representative solution

Let $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ be positions at t_1, t_2 after the known $\boldsymbol{r}_0$. The initial motion and applicable force corrections are chosen to minimize

$$J_r = \sum_{i=1}^{2} \|\boldsymbol{r}(t_i;\boldsymbol{\eta}) - \boldsymbol{p}_i\|^2, \qquad \boldsymbol{\eta}_{\text{M4}} = (\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L). \tag{18}$$

The next selection criterion favors the published force scale,

$$J_\alpha = (\alpha_D - 1)^2 + (\alpha_L - 1)^2, \tag{19}$$

using only the terms present in each model. Among equal-quality M4 states, the smallest spin magnitude selects a representative state. Deterministic multi-start optimization applies these criteria in order. The position-error allowance is 6×10^{-12} m^2, and the coefficient-objective allowance is 10^{-8}; both are numerical selection tolerances. Initial guesses use measured displacement velocities and the literature's experimental spin rates. M3 represents its spin axis by two angles; its unit vector serves only to carry direction in Eq. (8).

4.2.2 Comparison with recorded match flights

The Sportec Open DFL dataset associates optical tracking with match events [10, 11]. We analyze four flights from matches J03WOH and J03WQQ, and one recorded penalty. Events are selected for an identifiable shot and a tracked crossing of the actual goal-line plane. The four flights comprise two headers and two foot shots. Each inverse calculation uses the first three available positions at 0, 0.04 and 0.08 s. Later tracking coordinates determine the observed crossing of the goal plane. The penalty segment begins with a slower first displacement before the faster recorded flight; its early coordinates are used as recorded.

Table 7: All five recorded flights and all four inverse models. Error is the Euclidean difference in goal-plane (y, z).

Event	Model	Termination	Early residual (m)	Plane error (m)
WOH 0487 (head)	M1	plane	0.036	0.717
WOH 0487 (head)	M2	plane	0.00501	2.348
WOH 0487 (head)	M3	plane	2.28×10^{-12}	1.909
WOH 0487 (head)	M4	ground	3.53×10^{-10}	—
WOH 0606 (foot)	M1	plane	0.0263	4.988
WOH 0606 (foot)	M2	ground	0.0303	—
WOH 0606 (foot)	M3	ground	1.32×10^{-12}	—
WOH 0606 (foot)	M4	ground	1.83×10^{-10}	—
WOH 0968 (foot)	M1	ground	0.124	—
WOH 0968 (foot)	M2	ground	0.124	—
WOH 0968 (foot)	M3	early termination	0.127	—
WOH 0968 (foot)	M4	plane	0.126	10.893
WQQ 1227 (head)	M1	plane	0.18	2.785
WQQ 1227 (head)	M2	plane	0.172	3.325
WQQ 1227 (head)	M3	fit failure	0.0226	—
WQQ 1227 (head)	M4	fit failure	0.0952	—
Penalty	M1	plane	0.204	0.853
Penalty	M2	plane	0.204	0.853
Penalty	M3	fit failure	0.208	—
Penalty	M4	fit failure	0.208	—

Table 8: Reached-case errors and termination counts. Individual errors are retained in the preceding table.

Model	Reached	Ground	Other	Fit failures	Mean error (m)
M1	4/5	1	0	0	2.336
M2	3/5	2	0	0	2.175
M3	1/5	1	1	2	1.909
M4	1/5	2	0	2	10.893

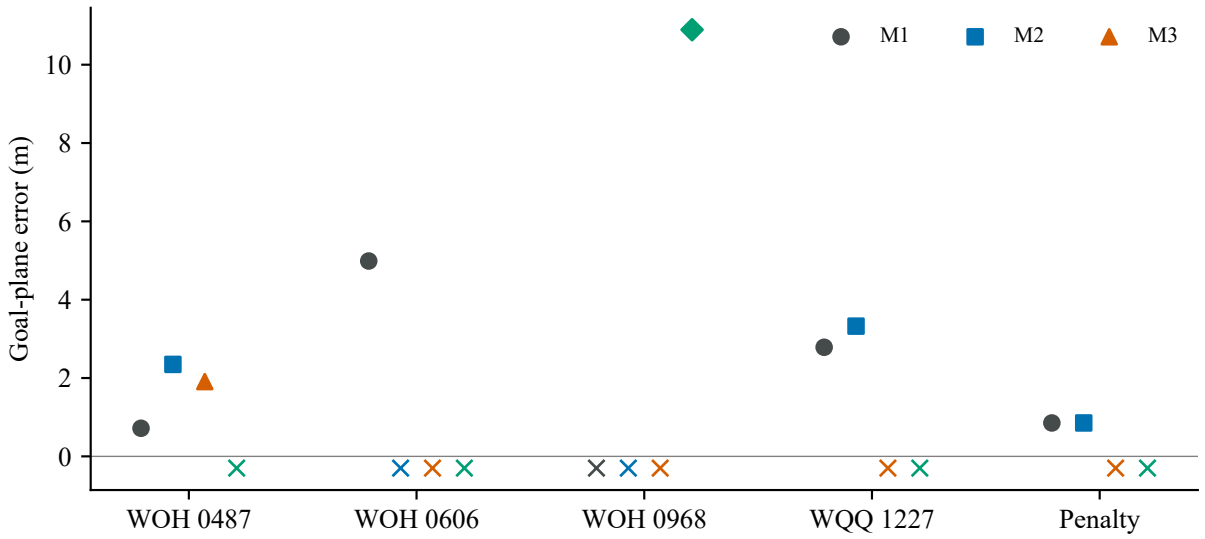


Figure 10: Individual goal-plane prediction errors for the five recorded flights. Each marker is one event and model. Crosses below zero denote trajectories that provide no goal-plane prediction; their termination status appears in Table 7.

M1 reaches the goal plane in 4 of five cases; M2 reaches the goal plane in 3 of five cases; M3 reaches the goal plane in 1 of five cases; M4 reaches the goal plane in 1 of five cases. M1 gives the largest number of goal-plane predictions in these five recorded flights. The table reports each reached-case mean together with its contributing count. Large force corrections and iteration-limit cases reveal a strong dependence on the early recorded motion. Fit failures denote numerical iteration limits; their candidate trajectories are retained in the calculation records.

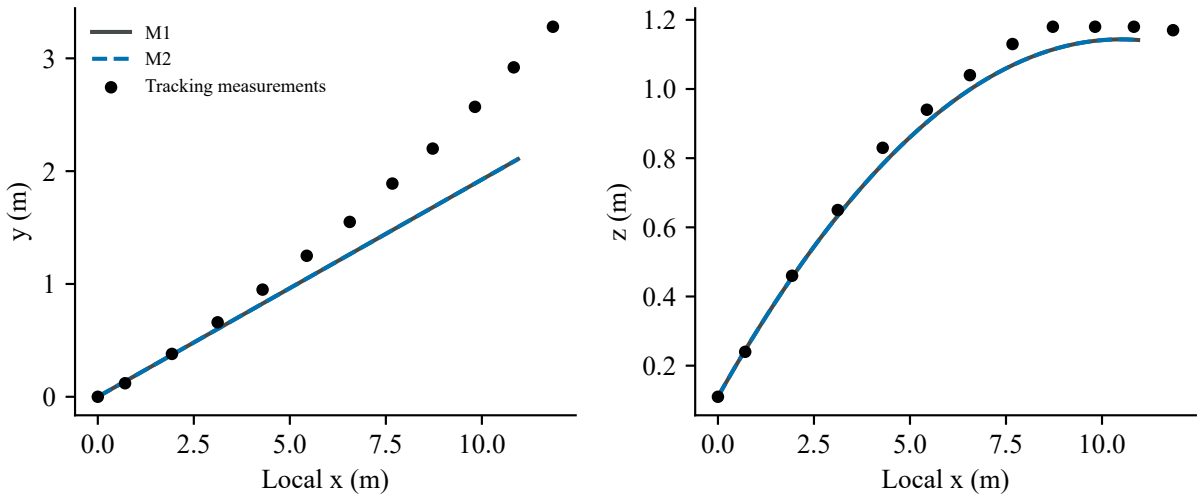


Figure 11: Converged inverse predictions for the recorded penalty, with all tracked positions shown. The first three positions enter the fit; remaining positions display the observed continuation. All model statuses are listed in Table 7.

The comparison links early acceleration to later flight. A large fitted drag correction produces rapid momentum loss, while lift corrections alter the direction of curvature. Changes in early recorded displacement therefore have a strong effect on the predicted endpoint. The event-level results show how each model responds to the actual recorded segment.

4.2.3 Competition Problem B

The supplied observations are

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

(20)

Applying the same inverse procedure gives Table 9. The preferred M4 initial conditions are

$$\begin{aligned} \boldsymbol{v}_0 &= (30.406, 9.426, 6.906)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (-49.090, -31.483, -112.266)\ \mathrm{rad/s}, \\ (\alpha_D, \alpha_L) &= (1.0001, 1.0000). \end{aligned} \tag{21}$$

Table 9: Inverse predictions for the competition observations. Residual is the maximum absolute early-coordinate difference.

Model	α_D	α_L	Residual (m)	y_g (m)	z_g (m)	t_g (s)
M1	—	—	0.016	3.217	1.957	0.3706
M2	1.5683	—	0.01	3.217	1.879	0.4058
M3	1.5666	1.0000	1.5×10^{-12}	2.661	1.938	0.4008
M4	1.0001	1.0000	6.4×10^{-13}	2.683	1.959	0.4026

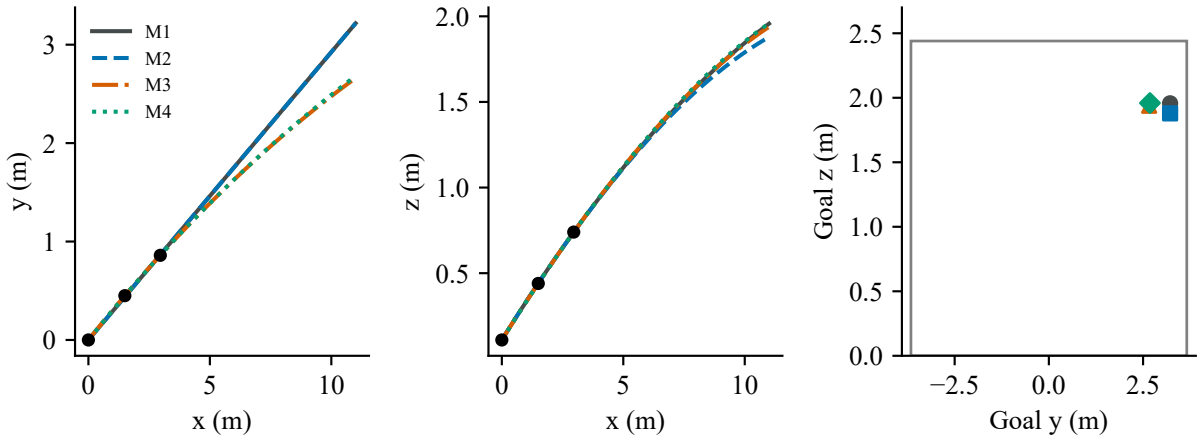


Figure 12: Four inverse predictions for Problem B: horizontal path, vertical path and ball-centre positions at the goal plane. Black points are the supplied early observations.

M1 and M2 predict nearly the same horizontal direction because gravity and drag preserve the horizontal velocity ratio. Adding Magnus lift in M3 and M4 bends the ball laterally. The M4 goal-plane centre is $(2.683, 1.959)$ m at 0.4026 s and fits inside the goal with the ball-radius margin. Its early maximum coordinate residual is 6.4×10^{-13} m. The inferred initial spin is 126.51 rad/s; it supplies the aerodynamic curvature required by this representative fit.

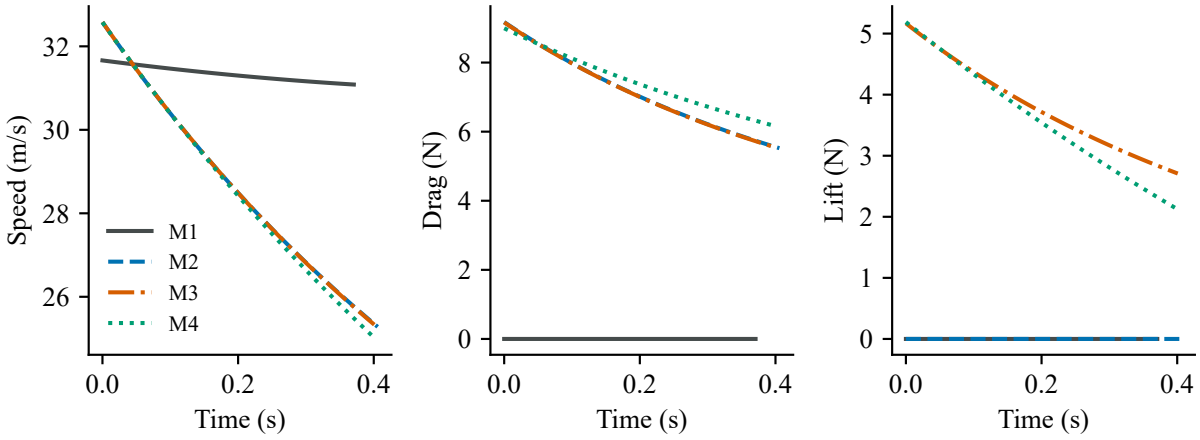


Figure 13: Speed and aerodynamic force magnitudes for the four fitted Problem B models. All share the same observations, while their force assumptions produce different continuation paths.

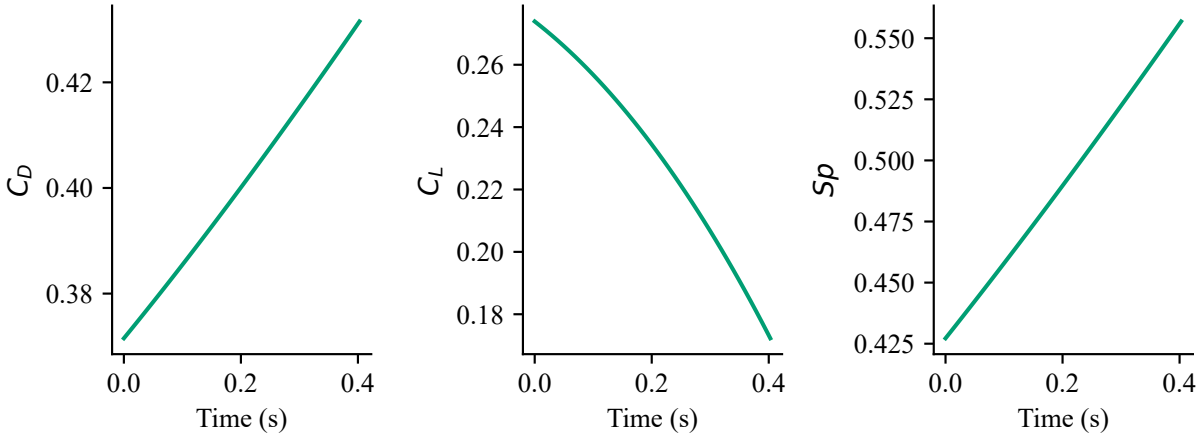


Figure 14: The complete M4 aerodynamic coefficients and spin parameter during the Problem B flight. Constant spin combined with decreasing speed raises Sp; the coupled empirical relations determine the resulting drag and lift coefficients.

The coefficient evolution in Figure 14 shows why constant spin still produces changing aerodynamics. The surface-to-air speed ratio increases as the ball slows, while the drag-dependent scaling also changes the lift coefficient.

4.3 Case Studies: Different Penalty-Kick Techniques

What happens when a goalkeeper faces not one textbook kick but a repertoire of different techniques? We examine five illustrative situations using the same three early positions at 0, 0.05 and 0.10 s. Each reference flight is generated with M4; its first three positions feed M1 and M4 with zero assumed spin. The figures show the generated flight, the zero-spin M4 continuation and

the input points. The chosen values describe examples, not statistical means or individual players [6, 9, 7, 8]. No observation noise or wind is added.

Case 1: Power Shot

The ball leaves at 27 m/s with an 8° elevation and zero spin. It reaches the goal quickly; in this example the M1 goal-plane error is 0.035 m, and the zero-spin M4 result is within 0.001 m. A goalkeeper has little time to react, although the weakly curved path is relatively straightforward under these assumed conditions.

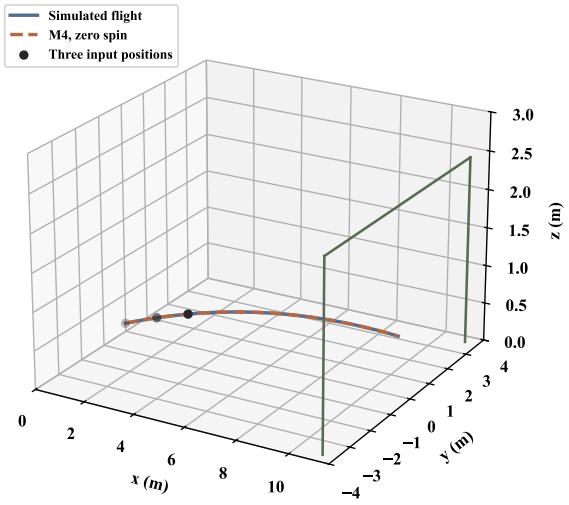


Figure 15: Case 1: power shot.

Case 2: Placement Shot

For a more controlled shot, we choose 20 m/s, an 11° elevation and zero spin. The simulated ball crosses the goal plane at a height of 0.617 m. M1 has a 0.071 m error, while zero-spin M4 is within 0.001 m. A lower speed leaves more flight time, so the goalkeeper can use the developing path to refine the expected position.

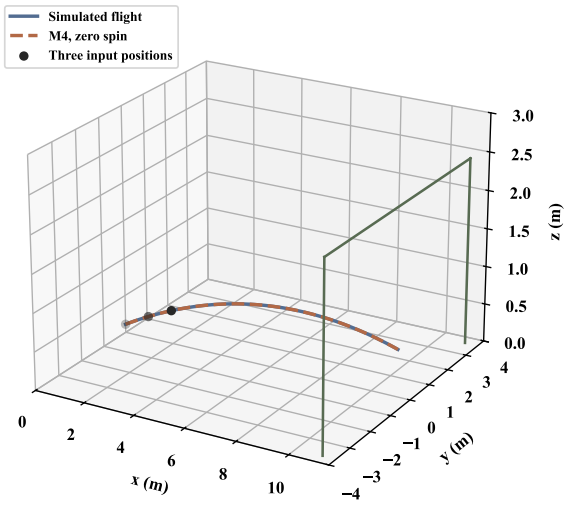


Figure 16: Case 2: placement shot.

Case 3: Curved Shot

The curved kick uses 24 m/s, a 10° elevation and vertical-axis spin $(0, 0, 35)$ rad/s. The flight bends sideways: its simulated goal crossing is $y = 0.892$ m, $z = 0.882$ m. When the prediction assumes zero spin, M4 misses this crossing by 0.727 m. The first positions alone can make the sideways motion appear partly attributable to initial direction, so a goalkeeper should allow for subsequent curvature.

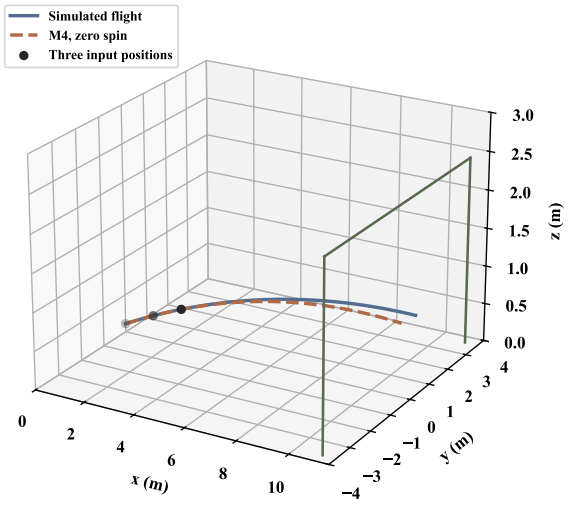


Figure 17: Case 3: sidespin curve.

Case 4: Topspin and Backspin

Both examples use 24 m/s. The topspin case takes 12° and $(0, 35, 0)$ rad/s; the backspin case takes 10° and $(0, -35, 0)$ rad/s. Their simulated goal heights are 0.382 m and 1.778 m, respectively. Zero-

spin M4 incurs errors of 0.750 m and 0.698 m. Opposite spin changes the vertical path, making height an important part of the goalkeeper's assessment.

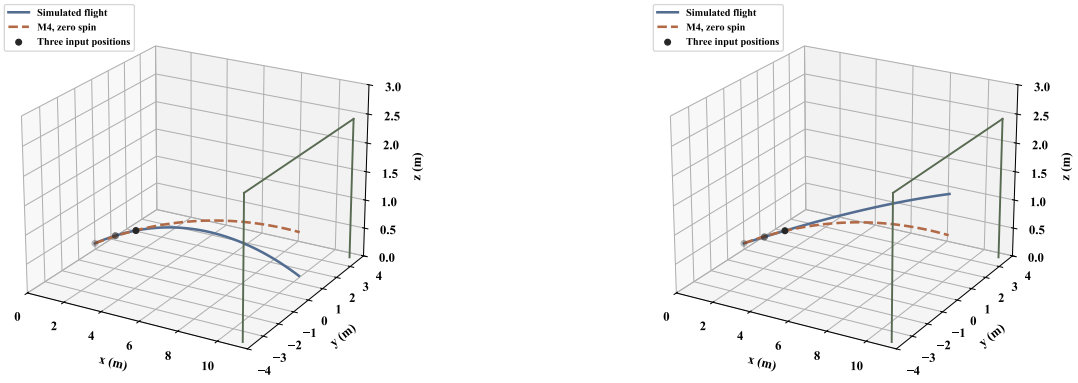


Figure 18: Case 4: topspin (left) and backspin (right).

Case 5: Chip Shot

The chip uses 14 m/s, a 20° elevation and $(0, -15, 0)$ rad/s backspin. Its higher launch angle and lower speed produce a longer, arcing flight; the simulated goal height is 0.368 m. M1 and zero-spin M4 have errors of 0.305 m and 0.086 m, respectively. Unlike a power kick, the goalkeeper must account for a slower ball that rises before descending.

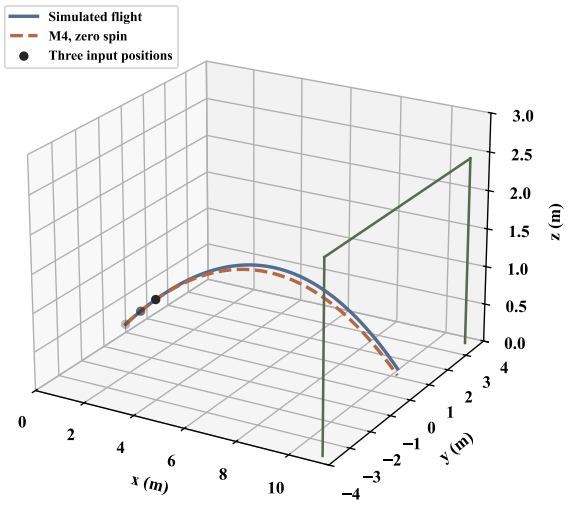


Figure 19: Case 5: chip shot.

Comparison. Table 10 collects the six trajectories. M4 with the generating spin supplied is a conditional control, not a three-position-only prediction: its near-zero error follows from using the same equations to generate and predict the trajectory. These single synthetic examples illustrate possible effects of spin; the recorded-match comparison remains a separate evaluation. Coefficients are fixed in these cases, unlike the multi-parameter inverse procedure above.

Table 10: Summary of illustrative three-point cases; goal-plane errors in metres

Case	v_0 (m/s)	Angle (deg)	Spin (ω_y, ω_z) (rad/s)	M1 (m)	M4 zero (m)	M4 known (m)
Power	27	8	$(0, 0)$	0.035	< 0.001	< 0.001
Placement	20	11	$(0, 0)$	0.071	< 0.001	< 0.001
Curved	24	10	$(0, 35)$	0.730	0.727	< 0.001
Topspin	24	12	$(35, 0)$	0.797	0.750	< 0.001
Backspin	24	10	$(-35, 0)$	0.651	0.698	< 0.001
Chip	14	20	$(-15, 0)$	0.305	0.086	< 0.001

5 Sensitivity Analysis

To be completed.

6 Discussion of Other Factors

We now examine surface geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [14] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



Figure 20: Illustrative comparison of a paneled football and a smooth sphere, inspired by Paudel and Yan [14]. Their compared bodies had diameter 0.22 m and mass 0.436 kg. The panel geometry shown here is schematic.

Alam et al. [13] measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere. Figure 21 shows the panels and surface textures of these footballs, while Figure 22 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.



Fig. 1. Vintage and Adidas Fevernova balls



Fig. 2. Adidas Teamgeist 2006 and 2009 balls



Fig. 3. Adidas 2010 Jabulani and Nike 2010 T-90 balls

Figure 21: Six footballs and details of their surfaces, reproduced from Figures 1–3 of Alam et al. [13].

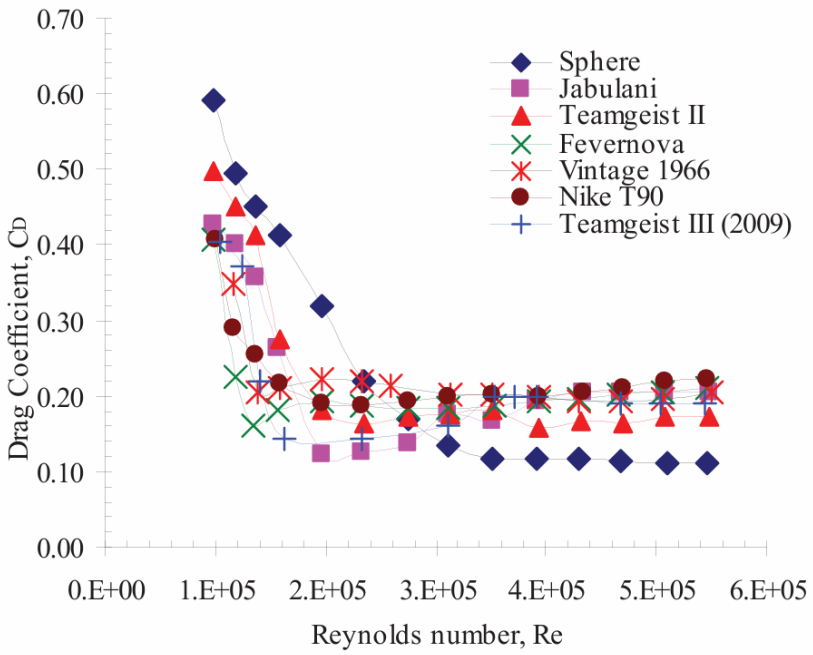


Figure 22: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [13].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [18], Supplementary Section 8.4, and the MuJoCo ellipsoid model [19], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{22}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [19]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\,Sp}. \tag{23}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{24}$$

It is 9.8×10^{-5} at the fitted competition launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [12], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_B + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{25}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

We repeat the M4 Problem B flight with exactly its fitted initial state and force corrections, first with constant spin and then with Eq. (25). The state is held fixed so the comparison measures the damping effect directly.

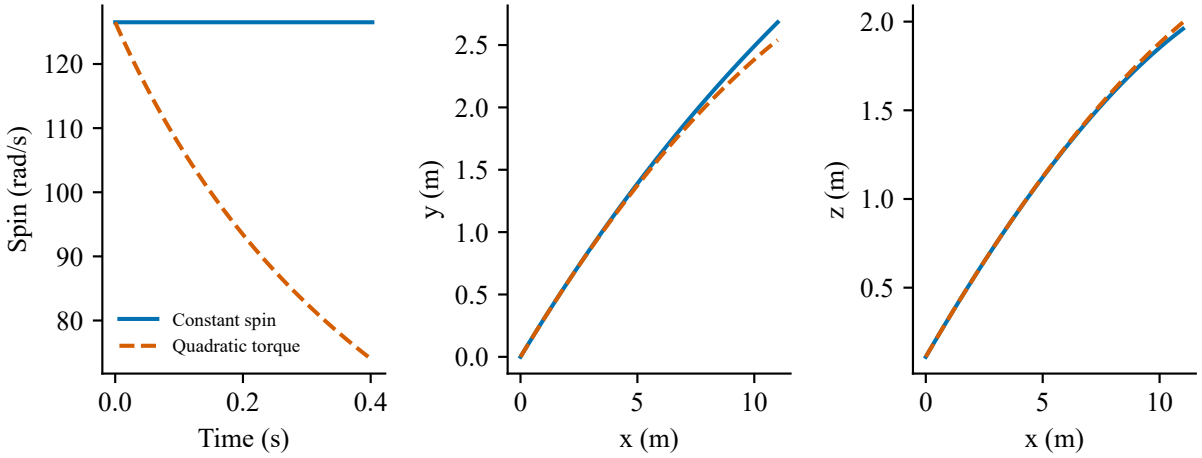


Figure 23: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for Problem B. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped goal-plane centre is $(2.541, 1.996)$ m, compared with $(2.683, 1.959)$ m at constant spin. The displacement is 0.147 m, or 1.34% of the 11 m flight distance. Spin decreases by 41.24%, and the goal-crossing time changes by -1.60%. Adding the much weaker viscous term changes the damped crossing by 6.5e-06 m. Thus this reference torque has a visible effect on spin and a smaller, measurable effect on the short-flight endpoint.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [12]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{26}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 11 evaluates these rough estimates at the fitted initial speed. The listed wind speeds are illustrative conditions for comparison.

Table 11: Illustrative wind estimates at the fitted launch speed. Percentages describe dynamic pressure at fixed aerodynamic coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Airflow angle (°)
1	+6.23	-6.05	+0.09	1.76
3	+19.27	-17.57	+0.85	5.26
5	+33.06	-28.34	+2.36	8.73

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.80$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The model relates soccer-ball flight to gravity, buoyancy, drag and Magnus lift; the numerical results presented here use the three-force baseline without buoyancy. The complete Teamgeist coefficient relations represent flow-state and spin changes, while the matched 2007 wind-tunnel comparison gives a closer reproduction for drag than for lift.

The controlled nonspinning example shows that including drag is essential for range prediction. Its M4 range is 21.871 m against the published 21.9 m. The rotating trajectory segments display the downward curvature of topspin and variation in the accuracy of the common coefficient law. Recorded match comparisons show how early displacement and fitted force corrections affect subsequent prediction.

For Problem B, M4 gives $(y, z) = (2.683, 1.959)$ m at the goal plane. Magnus lift accounts for the marked horizontal difference from the gravity-only and drag-only predictions. The same-state rotational-damping comparison shifts the crossing by 0.147 m under the documented torque setting. Wind estimates give the weak-wind condition for the still-air approximation. Sensitivity-analysis findings will be added after Section 5 is completed.

8 Model Evaluation

8.1 Physical Interpretation and Computational Use

The hierarchy makes the contribution of each force visible. The same dynamics handles forward calculation, early-position inversion and endpoint prediction. Force corrections provide a direct way to accommodate measured trajectory differences, and all experimental and numerical inputs are retained with their sources.

8.2 Predictive Performance and Improvements

The wind-tunnel comparisons, controlled launches and recorded match flights examine complementary aspects of the model: coefficient trends, curvature and range, and prediction from early observations. Their event-level errors identify where a common football coefficient law and an early trajectory segment give a useful continuation. Ball-specific aerodynamic measurements, directly measured spin and synchronized higher-rate positions would improve this connection. Explicit surface orientation, rotational damping and wind can be included as the application requires.

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Reproduction Code

The accompanying project contains the frozen inputs, vector-marker extraction, four-model calculation, result tables and figure scripts. The following excerpts show the shared state equation and force evaluation. The inverse solver uses the same state derivative with the three selection criteria described in Section 4.2. Calculation and figure generation are separate commands, allowing plots to be refreshed from saved results.

```
--
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
--
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
        0.])
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```